

LINUX PROGRAMMING LAB

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS18	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

OBJECTIVES:

The course should enable the students to:

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols.

COURSE OUTCOMES:

After completion of the course Students is able to:

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Environment using various protocols.

LIST OF EXPERIMENTS

1. Linux Commands & Introduction of Shell
2. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
3. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
4. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
5. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.
6. Write a shell script that accepts a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
7. Write a shell script to list all of the directory files in a directory.
8. Write an awk script to count the number of lines in a file that do not contain vowels.
9. Write an awk script to find the number of characters, words and lines in a file.
10. Write a c program that makes a copy of a file using standard I/O and system calls.
11. Implement in C the following UNIX commands using System calls.
 - i. A. cat B. mv
12. Write a C Program to list files in a directory.
13. Write a C program to list for every file in a directory, its inode number and file name.
14. Write a C program to create a child process and allow the parent to display "parent" and the child to display "child" on the screen.
15. Write a C program to create a Zombie process.
16. Write a C program that illustrates how an orphan is created.
17. Write C programs that illustrate communication between two unrelated processes using named pipe(FIFO).
18. Write a C program in which a parent writes a message to a pipe and the child reads the message.
19. Write Client server program in C for connection oriented communication between server and client process using unix domain sockets to perform the following
20. Write Client server program in C for connection oriented communication between server and client process using internet domain sockets.

TEXT BOOKS:

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| 1. Unix System Programming using C++, T.Chan, PHI. |
| 2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH,2006. |

REFERENCES:

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| 1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008 |
| 2. Linux System Programming, Robert Love, O'Reilly, SPD. |
| 3. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education. |
| 4. Unix Network Programming ,W.R.Stevens,PHI. |
| 5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education. |